

PH 513: Quantum Mechanics I

Course Code	PH 513
Course Name	Quantum Mechanics I
L-T-P-C	3-0-0-3
Prerequisites	Undergraduate core physics courses
Intended for	Core for M.Sc. and I-Ph.D. in Physics; Elective for B.Tech. 3rd and 4th Years.
Mutual Exclusion	None
Approval	9th Senate (May 2016), in the Minor in Physics curriculum
Revised	64th BoA (May 2026) - syllabus revised

Preamble

This course is an introductory postgraduate level course on quantum mechanics covering its basic principles. Several applications of quantum mechanics will be discussed to train students to apply these ideas to model systems in both one and higher dimensions. The course begins with a review discussion of Schrodinger formalism complemented with applications. Dirac notation will be introduced. Applications of quantum mechanics in three dimensions will be discussed. Approximation techniques such as WKB and time-independent perturbation theory will be also discussed in this course.

Course Modules

- **Unit/Topic 1:** Review of Schrodinger formalism with applications in 1D, Dirac notation, linear vector operators, matrix representation of states and operators, Ehrenfest theorem, commutator relations in quantum mechanics and uncertainty relations (12 Hours)
- **Unit/Topic 2:** Theory of angular momentum in quantum mechanics, commutator relations in angular momentum, Eigen values and Eigen states of angular momentum, spin-angular momentum. (9 hours)
- **Unit/Topic 3:** Application of Schrodinger equation in 3-D models, symmetry and degeneracy, central potentials, Schrodinger equation in spherical co-ordinates, solution to hydrogen atom problem. (9 hours)
- **Unit/Topic 4:** Schrodinger and Heisenberg pictures, Heisenberg equation of motion, Interaction picture, WKB method, Time independent non-degenerate and degenerate perturbation theory with applications. Fine structure, Zeeman effect and hyperfine splitting of one-electron systems (12 hours)

Laboratory/practical/tutorial Modules: No

Text Books

1. D J Griffith, *Introduction to quantum mechanics* (Pearson, Fourth edition, 2015).
2. Claude Cohen-Tannoudji, B Diu, F Laloe, *Quantum Mechanics – Vol.1*, (Wiley, Second edition 2020).
3. R Shankar, *Principles of Quantum Mechanics* (Plenum Press, Second edition, 2011)

References

1. *Modern Quantum Mechanics* – J J Sakurai (Pearson Edn., revised edition, 1994)
2. *Quantum Mechanics Concepts and Applications* – N Zetilli (2nd edition 2009)
3. *The Feynman Lectures in Physics*, Vol. 3, R.P. Feynman, R.B. Leighton, and M. Sands (Narosa Publishing House, 1992).
4. *Practical Quantum Mechanics* – Siegfried Flügge (Springer 1994)
5. *Quantum Mechanics* – P. C. Deshmukh (Cambridge University Press 2023)